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MIL-M-28787/522(SH)
22 December 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SYNCHRONOUS SERIAL CONTROL MODULE, KEY CODE AFK

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic synchronous serial control module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.826 failures per million operating hours for class I or 7.816 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AFK.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A22	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B8	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	----------Module shall pass all remaining tests----------						
D1	Ibd	V	S	-0.12	1.11	-0.12	1.11	-0.12	1.11	mA
D2	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D3	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D4	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D5	Ibd	V	S	-0.12	1.21	-0.12	1.21	-0.12	1.21	mA
D6	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D7	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D8	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D9-D15	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D16	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D17	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	mA
D18-D24	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D25	Ibd	V	S	-0.12	0.32	-0.12	0.32	-0.12	0.32	mA
D26	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E5	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E6,E7	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E8-E10	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E11	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E12-E16	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E17-E24	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E25-E40	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F2	Iih	VII	X	-5.68	-2.86	-5.68	-2.86	-5.68	-2.86	mA
F3,F4	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F5	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F6-F12	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F13	Iih	VII	X	-12.7	-2.75	-12.7	-2.75	-12.7	-2.75	mA
F14	Iih	VII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
F15	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F16	Iih	VII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
F17-F21	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F22	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F23	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F24	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F25	Iih	VII	X	-8.0	0.2	-8.0	0.2	-8.0	0.2	mA
F26-F32	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F33	Iih	VII	X	-5.83	-2.83	-5.83	-2.83	-5.83	-2.83	mA
F34-F42	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F43-F58	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G1	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G2	Iil	VIII	X	-5.68	-2.86	-5.68	-2.86	-5.68	-2.86	mA
G3, G4	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G5	Iil	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G6-G12	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G13	Iil	VIII	X	-12.7	-2.75	-12.7	-2.75	-12.7	-2.75	mA
G14	Iil	VIII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
G15	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G16	Iil	VIII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
G17-G21	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G22	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G23	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G24	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G25	Iil	VIII	X	-8.0	0.2	-8.0	0.2	-8.0	0.2	mA
G26-G32	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G33	Iil	VIII	X	-5.83	-2.83	-5.83	-2.83	-5.83	-2.83	mA
G34-G42	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G43-G58	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
H1-H42	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J43	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K	Iozh	XI	X	0.0	1.0	0.0	1.0	0.0	1.0	mA
L1-L4	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L5-L9	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L10	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11-L13	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L14-L21	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M	PSFC	XIII	S	6.6	12.4	6.6	12.4	6.6	12.4	uF
N	Icc	XIV	X	510	1240	510	1240	510	1240	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R2	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R3	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R4	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R5	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R6,R7	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R8	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R9	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R10	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R11-R13	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R14	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R15,R16	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R17	Ci/o	XVIII	Q	8	100	8	100	8	100	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R18-R21	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R22	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R23,R24	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R25	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R26-R38	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R39	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R40,R41	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R42	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R43-R49	Ci/o	XVIII	Q	3	75	3	75	3	75	pF
R50,R51	Ci/o	XVIII	Q	3	100	3	100	3	100	pF
R52-R58	Ci/o	XVIII	Q	3	75	3	75	3	75	pF
T	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/522-1 for class I and M28787/522-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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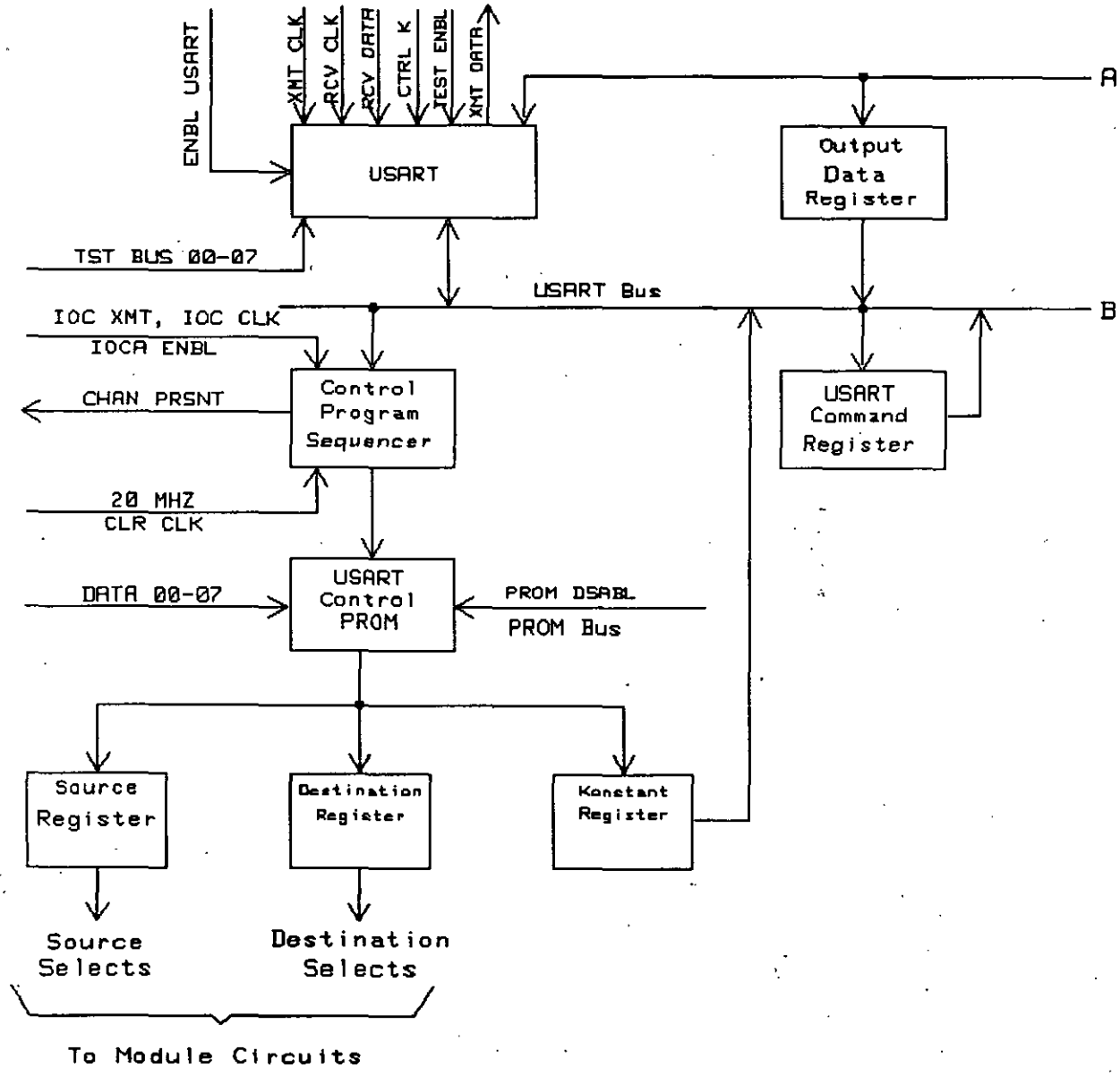


FIGURE 1. AFK functional block diagram.

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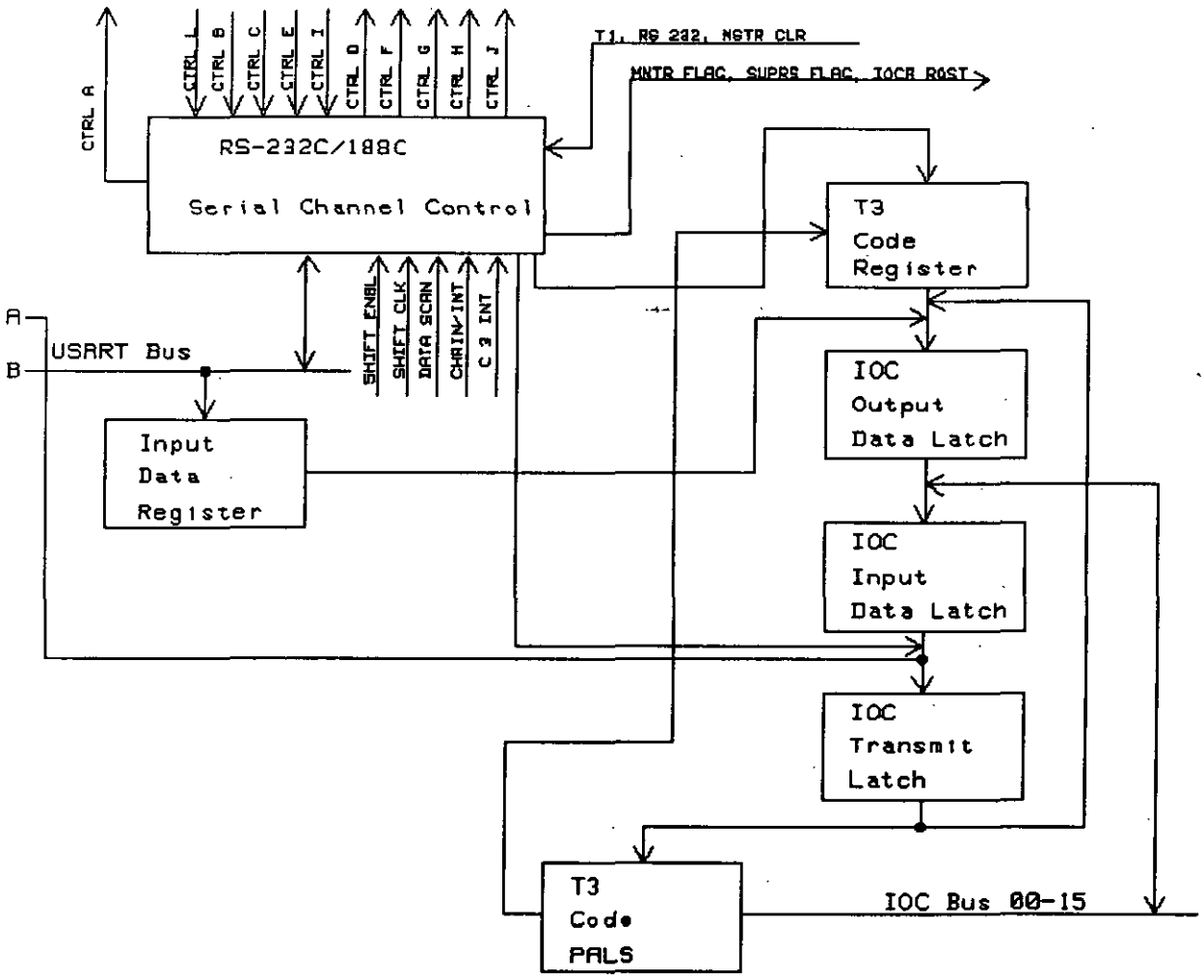


FIGURE 1. AFK functional block diagram - Continued.

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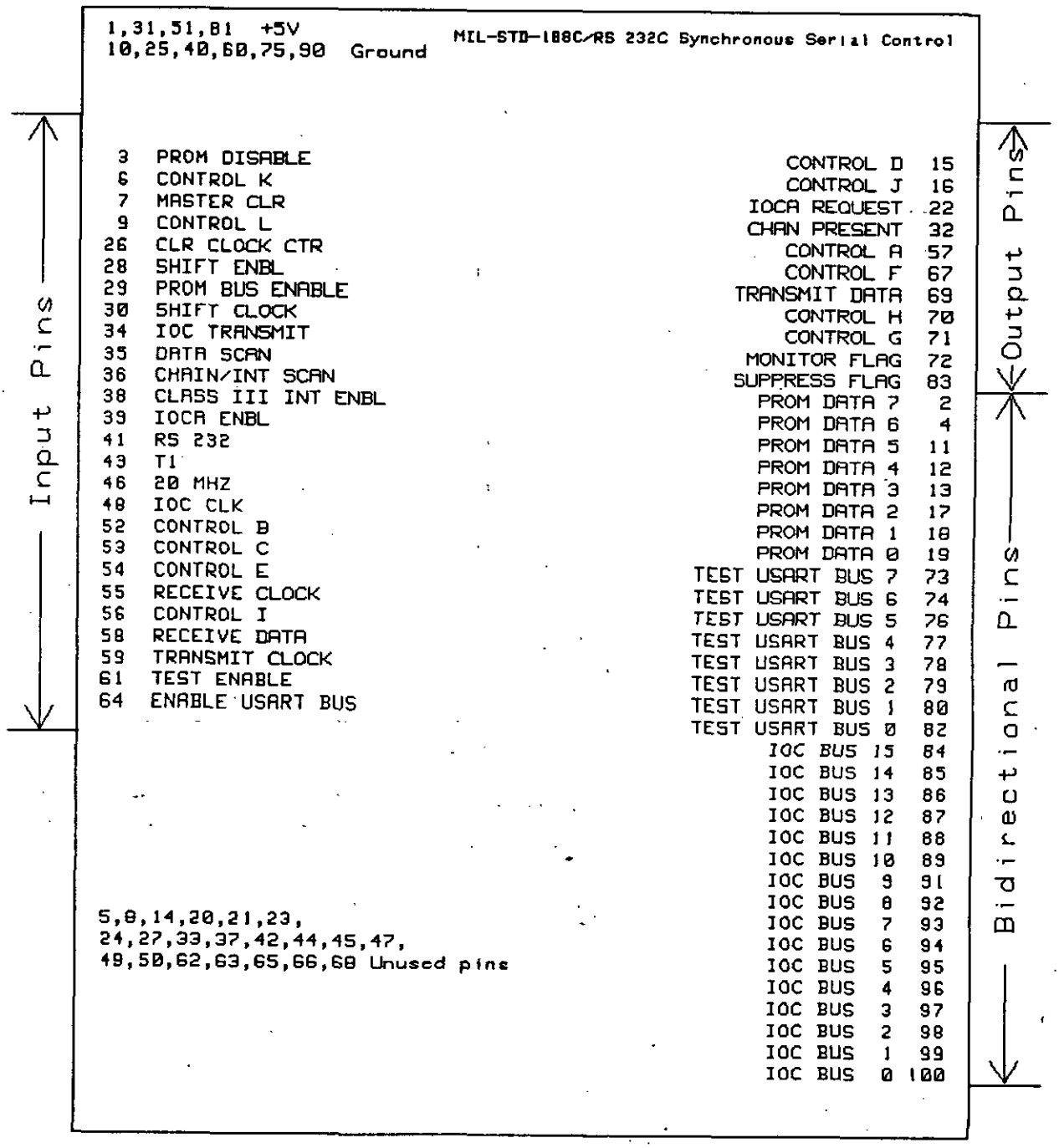


FIGURE 2. AFK pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

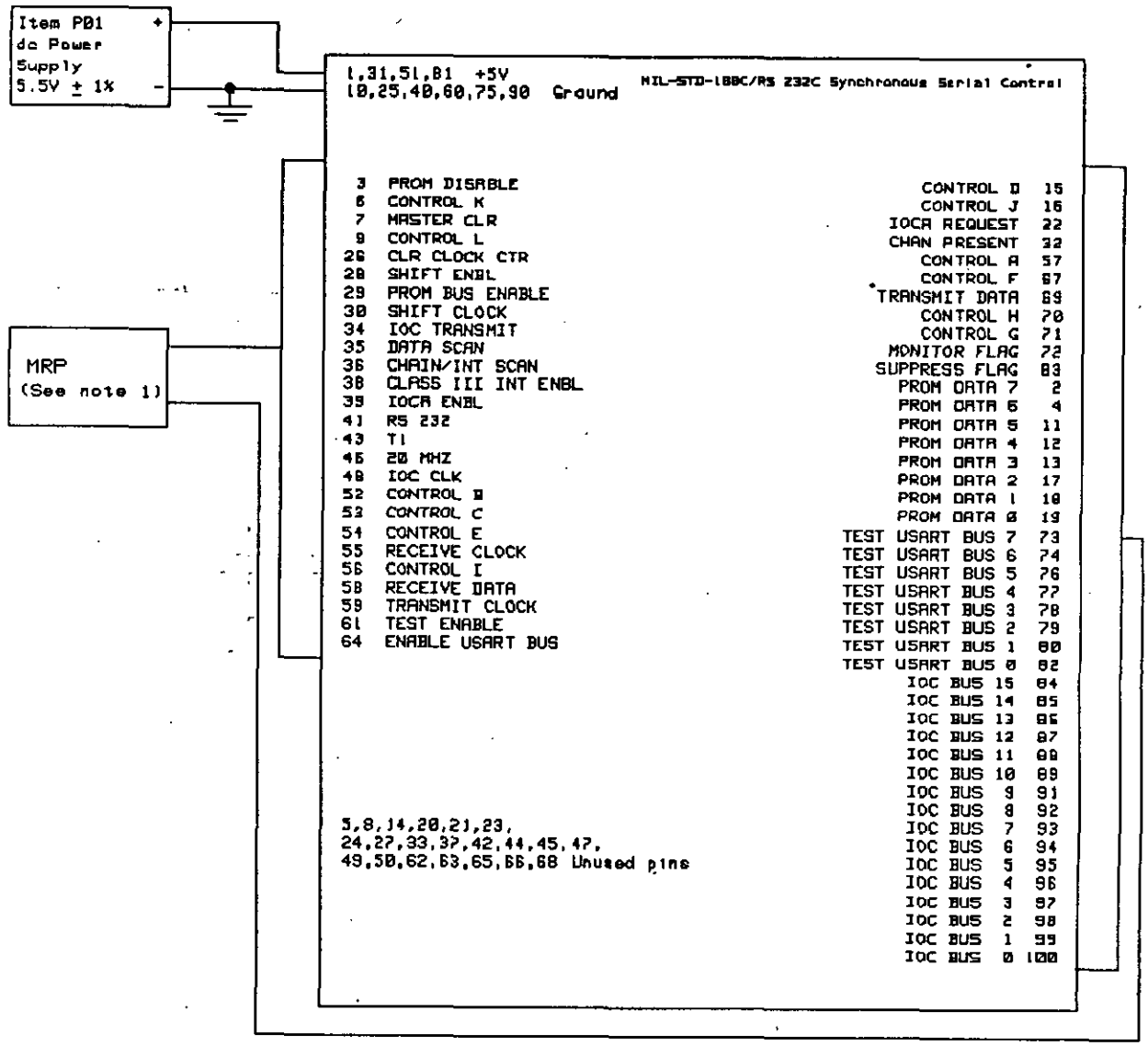
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AFK operating temperature and component temperature test circuit.

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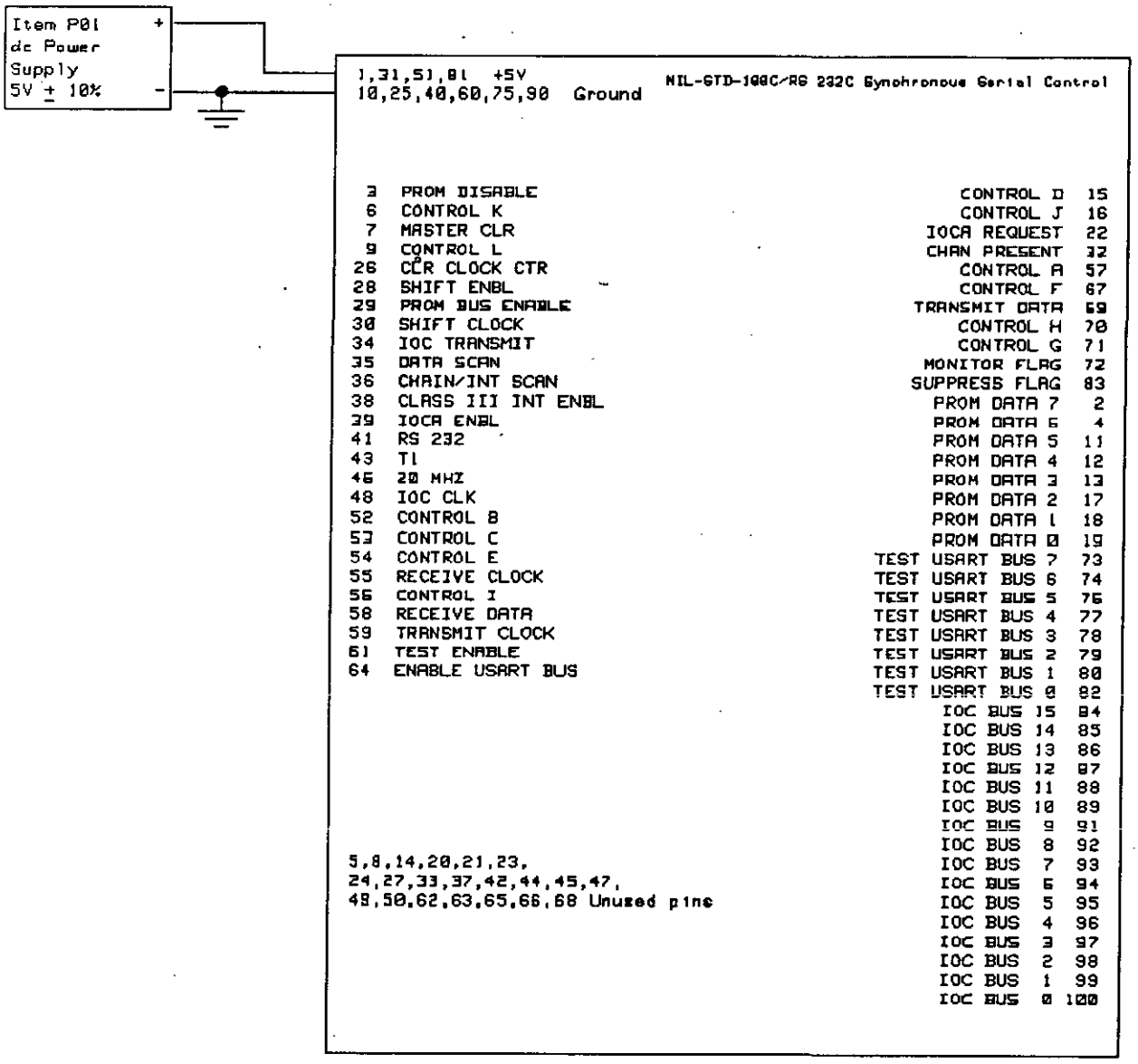


FIGURE 4. AFK vibration test circuit.

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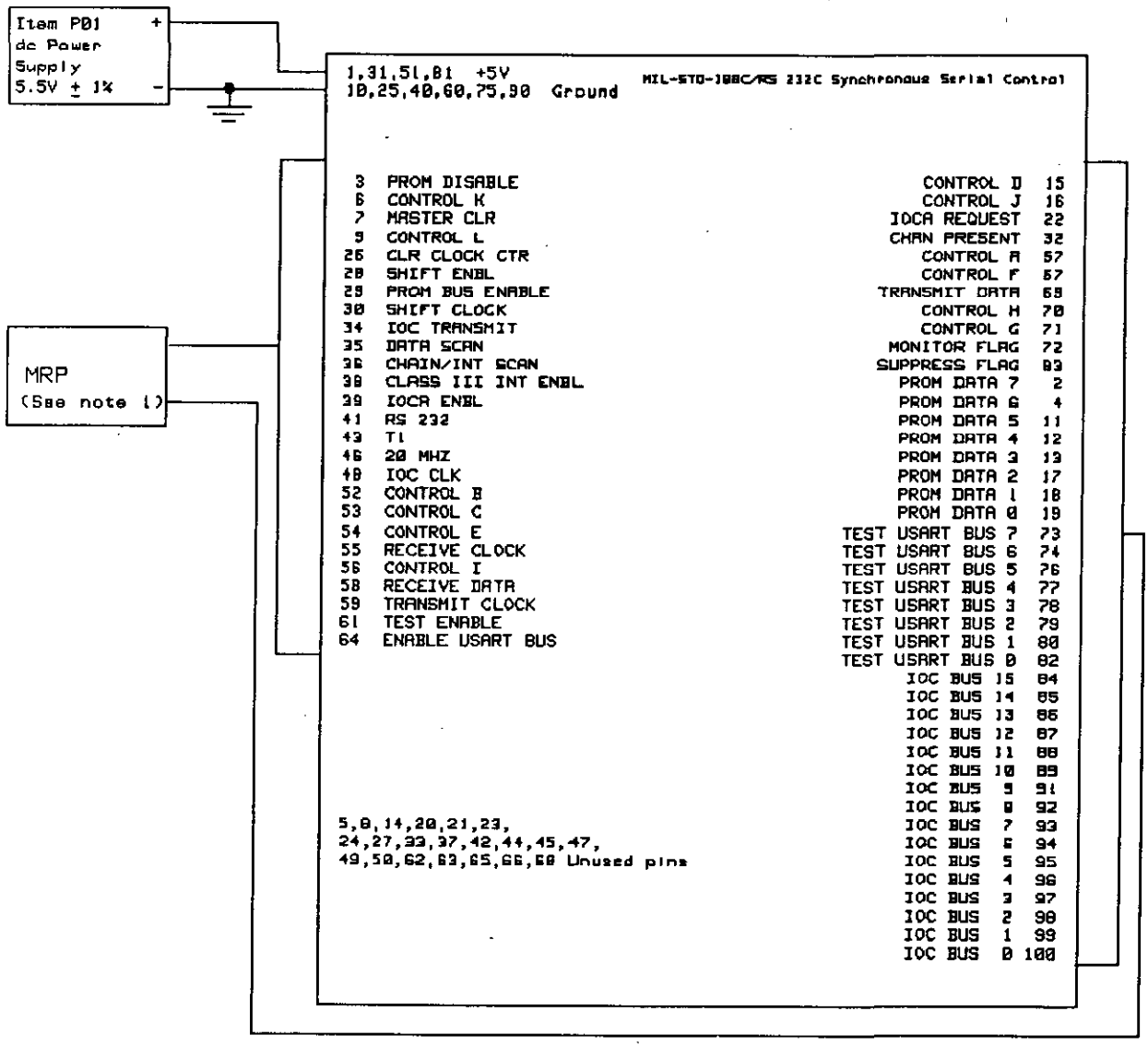
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AFK life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 6.82 W (max); 4.56 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PROM DSABL	NA	0.4	2.4	-5.68	-2.86	1.11	75
CTRL K	-1.5	0.4	2.4	-0.2	0.02	0.1	75
MSTR CLR	-1.5	0.4	2.4	-0.4	0.04	0.2	100
CTRL L	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CLR CLK	NA	0.4	2.4	-12.7	-2.75	1.21	100
SHIFT ENBL	NA	0.4	2.4	-6.03	-2.85	0.22	75
PROM BUS	-1.5	0.4	2.4	-0.2	0.02	0.1	75
SHIFT CLK	NA	0.4	2.4	-6.03	-2.85	0.22	100
IOC XMT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DATA SCAN	-1.2	0.4	2.4	-0.6	0.02	0.1	100
CHAIN/INT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
C 3 INT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOCA ENBL	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RS 232	-1.5	0.4	2.4	-0.2	0.02	0.1	100
T1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
20 MHZ	-1.2	0.4	2.4	-2.0	0.05	1.0	75
IOC CLK	-1.2	0.4	2.4	-8.0	0.2	4.0	100
CTRL B	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CTRL C	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CTRL E	-1.5	0.4	2.4	-0.2	0.02	0.1	75
RCV CLK	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CTRL I	-1.5	0.4	2.4	-0.2	0.02	0.1	50
RCV DATA	-1.5	0.4	2.4	-0.2	0.02	0.1	75
XMT CLK	-1.5	0.4	2.4	-0.2	0.02	0.1	75
TEST ENBL	NA	0.4	2.4	-5.83	-2.83	0.32	100
ENBL USART	-1.5	0.4	2.4	-0.2	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DATA 07	NA	0.4	2.4	-0.2	0.02	NA	75
DATA 06	NA	0.4	2.4	-0.2	0.02	NA	75
DATA 05	NA	0.4	2.4	-0.2	0.02	NA	75
DATA 04	NA	0.4	2.4	-0.2	0.02	NA	75
DATA 03	NA	0.4	2.4	-0.2	0.02	NA	75
DATA 02	NA	0.4	2.4	-0.2	0.02	NA	75
DATA 01	NA	0.4	2.4	-0.2	0.02	NA	75
DATA 00	NA	0.4	2.4	-0.2	0.02	NA	75
TST BUS 07	NA	0.4	2.4	-0.2	0.02	NA	75
TST BUS 06	NA	0.4	2.4	-0.2	0.02	NA	75
TST BUS 05	NA	0.4	2.4	-0.2	0.02	NA	75
TST BUS 04	NA	0.4	2.4	-0.2	0.02	NA	75
TST BUS 03	NA	0.4	2.4	-0.2	0.02	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
TST BUS 02	NA	0.4	2.4	-0.02	0.02	NA	75
TST BUS 01	NA	0.4	2.4	-0.02	0.02	NA	75
TST BUS 00	NA	0.4	2.4	-0.02	0.02	NA	100
IOC BUS 15	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 14	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 13	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 12	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 11	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 10	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 09	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 08	NA	0.4	2.4	-0.25	0.07	NA	100
IOC BUS 07	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 06	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC BUS 05	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 04	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 03	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 02	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 01	NA	0.4	2.4	-0.25	0.07	NA	75
IOC BUS 00	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CTRL D	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CTRL J	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IOCA RQST	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CHAN PRSNT	NA	0.5	NA	1.6	1.0	1.0	NA	NA
CTRL A	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CTRL F	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
XMT DATA	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CTRL H	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CTRL G	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MNTR FLAG	2.4	0.5	-0.16	1.6	NA	NA	NA	NA
SUPRS FLAG	2.4	0.5	-0.16	1.6	NA	NA	NA	NA
DATA 07	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
DATA 06	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DATA 05	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
DATA 04	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
DATA 03	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
DATA 02	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
DATA 01	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
DATA 00	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TST BUS 07	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TST BUS 06	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TST BUS 05	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TST BUS 04	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TST BUS 03	2.4	0.5	-0.16	1.6	NA	NA	-40.0	100
TST BUS 02	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
TST BUS 01	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
TST BUS 00	2.4	0.5	-0.16	1.6	NA	NA	-40.0	100
IOC BUS 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 09	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 08	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC BUS 07	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC BUS 06	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 05	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 04	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
IOC BUS 03	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 02	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 01	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC BUS 00	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	IOC CLK to IOC BUS 00	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 01	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 02	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 03	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 04	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 05	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 06	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 07	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 08	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 09	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 10	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 11	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 12	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 13	-	-	-	67	67	67	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	IOC CLK to IOC BUS 14	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 15	-	-	-	67	67	67	nS	12
2	tsu	IOC BUS 00 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 01 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 02 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 03 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 04 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 05 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 06 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 07 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 08 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 09 to IOC CLK	24	24	24	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tsu	IOC BUS 10 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 11 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 12 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 13 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 14 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 15 to IOC CLK	24	24	24	-	-	-	nS	12
3	tsu	IOC XMT to IOC CLK	30	30	30	-	-	-	nS	12
3	tsu	IOCA ENBL to IOC CLK	30	30	30	-	-	-	nS	12
4	tpd	20 MHZ to DATA 00	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 01	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 02	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 03	-	-	-	250	250	250	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	20 MHZ to DATA 04	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 05	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 06	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 07	-	-	-	250	250	250	nS	12
5	tpd	20 MHz to TST BUS 00	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 01	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 02	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 03	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 04	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 05	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 06	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 07	-	-	-	192	192	192	nS	12
6	tpd	20 MHZ to SUPRS FLAG	-	-	-	81	81	81	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tpd	DATA SCAN to IOCA RQST	-	-	-	91	91	91	nS	12
8	tpd	TEST ENBL to DATA 00	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 01	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 02	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 03	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 04	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 05	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 06	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 07	-	-	-	52	52	52	nS	12

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6.4 Module function. MIL-STD-188C/RS 232C/Variable Character Length Synchronous (VACALES) synchronous serial control module (AFK) and MIL-STD-188C/RS 232C asynchronous serial control module (AEK) direct serial Input/Output (I/O) operation for a single channel of a two-channel group. The following paragraphs describe both modules since their operations are similar. The synchronous module contains data latches, a Universal Synchronous Asynchronous Receiver Transmitter (USART) and associated control Programmable Read Only Memory (PROM), a control program sequencer, command register, source register, destination register and constant register, input and output data registers, a T3 code register and a serial channel control gate array. The asynchronous control module contains all of the logic listed for the synchronous module plus external suppress and monitor detection logic. The suppress and monitor logic for the synchronous module is contained within the USART.

6.4.1 USART and control PROM. The USART for the asynchronous control module is a Programmable Communication Interface (PCI) integrated circuit that accepts parallel input data, formats and serializes the information based on its current operating mode and transmits the data in a serial bit stream. Simultaneously, serial data can be received, converted into parallel form, deformatted and sent to the Input/Output Controller (IOC) via the IOC bus. Operation of both USARTs is directed by a microprogram contained in a control PROM. The control PROM is a 2K x 8-bit memory with outputs sent to the source, destination and constant registers. The control program sequencer addresses the PROM.

6.4.2 Control program sequencer. The control program sequencer directs asynchronous I/O operations by addressing the USART control PROM for the routine required to handle the condition in effect. USART bus bits are applied to an encoder which directs the upper counter to the value 6 through 0 (excluding 4). The enable sequencer signal from the channel control gate array provides an up-count enable to the lower counter. The carry-out function of the lower counter is applied to the counter for sequencer address bits 7 - 4 to allow sequential accessing through the microprogram. Address bits are applied to the control PROM.

6.4.3 Serial channel control gate array. The serial channel control gate array is an 84-pin leadless microcircuit designed to control serial I/O operations and interface the Input/Ouput Channel Adapter (IOCA) to the IOC.

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6.4.4 T3 code register. The T3 code register is an 8-bit register used to hold the T3 code from the serial channel control gate array. This code will be transmitted to the IOC at T3 time (via the IOC output data latch).

6.4.5 Data latches and transmit latch. The IOC output data latch, IOC input data latches and the IOC transmit latch are 16-bit, 16-bit and 8-bit latches, respectively. They are used to hold IOC input or output data. The IOC transmit latch receives data from the serial channel control gate array for transmission to the IOC.

6.4.6 Input and output data registers. The input data registers and output data register transfer data to and from the USART bus. The output data register is enabled by the source register bit 0 and clocked by the serial channel control gate array. The input data register is clocked by destination register bit 7 and enabled by the serial channel control gate array.

6.4.7 Command register. The USART command register is loaded by destination register bit 6 and its output enabled by source register bit 1. The command register is loaded from the serial channel control gate array or constant register via the USART bus, with outputs going to the control sequencer and being returned to the USART bus.

6.4.8 Source, destination and constant registers. The source, destination and constant are 8-bit registers used to hold control PROM outputs. The contents of odd control PROM addresses are sent to the source register. The contents of even control PROM addresses are sent to the destination register to the constant register depending upon the condition of source register bit 6. The constant register holds an 8-bit constant that is applied to the USART bus.

6.4.9 Monitor register. The monitor register is an 8-bit register that holds a character to be compared against an incoming bit stream. When the received character matches the one in the monitor register, the monitor flag to the serial control gate array is set. The monitor register is loaded from the IOCA bus by the serial channel control gate array as a result of an IOC execution of a load control memory macroinstruction (I/O control memory location 9). The monitor register on the synchronous control module is contained within the USART gate array.

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6.4.10 Suppress register. The suppress register is an 8-bit register that holds a character which is to be ignored by the IOC. When a character in an incoming bit stream matches the contents of the suppress register, the suppress flag is generated to the serial channel control gate array and that character is ignored. The suppress register is loaded from the IOCA bus by the serial channel control gate array as the result of an IOC execution of the load control memory macroinstruction (I/O control memory location 8). The suppress register on the synchronous control module is contained within the USART gate array.

6.4.11 Monitor and suppress comparators. The monitor and suppress comparators are 8-bit comparators that are enabled by the serial channel control gate array as the result of an IOC execution of a set monitor or set suppress (chaining) macroinstruction. The outputs from the comparators are the monitor and suppress flags. The monitor and suppress comparators on the synchronous control module are contained within the USART gate array.

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6.4.12 Description of I/O pins.

Signal name	Pin number	I/O	Description
PROM DSABL	3	I	PROM DISABLE
CTRL K	6	I	CONTROL K
MSTR CLR	7	I	MASTER CLR
CTRL L	9	I	CONTROL L
CLR CLK	26	I	CLR CLOCK CTR
SHIFT ENBL	28	I	SHIFT ENBL
PROM BUS	29	I	PROM BUS ENABLE
SHIFT CLK	30	I	SHIFT CLOCK
IOC XMT	34	I	IOC TRANSMIT
DATA SCAN	35	I	DATA SCAN
CHAIN/INT	36	I	CHAIN/INT SCAN
C 3 INT	38	I	CLASS III INT ENBL
IOCA ENBL	39	I	IOCA ENBL
RS 232	41	I	RS 232
T1	43	I	T1
20 MHZ	46	I	20 MHZ
IOC CLK	48	I	IOC CLK
CTRL B	52	I	CONTROL B
CTRL C	53	I	CONTROL C
CTRL E	54	I	CONTROL E

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6.4.12 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RCV CLK	55	I	RECEIVE CLOCK
CTRL I	56	I	CONTROL I
RCV DATA	58	I	RECEIVE DATA
XMT CLK	59	I	TRANSMIT CLOCK
TEST ENBL	61	I	TEST ENABLE
ENBL USART	64	I	ENABLE USART BUS
CTRL D	15	O	CONTROL D
CTRL J	16	O	CONTROL J
IOCA RQST	22	O	IOCA REQUEST
CHAN PRSNT	32	O	CHAN PRESENT
CTRL A	57	O	CONTROL A
CTRL F	67	O	CONTROL F
XMT DATA	69	O	TRANSMIT DATA
CTRL H	70	O	CONTROL H
CTRL G	71	O	CONTROL G
MNTR FLAG	72	O	MONITOR FLAG
SUPRS FLAG	83	O	SUPPRESS FLAG
DATA 07	2	I/O	PROM DATA 7
DATA 06	4	I/O	PROM DATA 6
DATA 05	11	I/O	PROM DATA 5

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6.4.12 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DATA 04	12	I/O	PROM DATA 4
DATA 03	13	I/O	PROM DATA 3
DATA 02	17	I/O	PROM DATA 2
DATA 01	18	I/O	PROM DATA 1
DATA 00	19	I/O	PROM DATA 0
TST BUS 07	73	I/O	TEST USART BUS 7
TST BUS 06	74	I/O	TEST USART BUS 6
TST BUS 05	76	I/O	TEST USART BUS 5
TST BUS 04	77	I/O	TEST USART BUS 4
TST BUS 03	78	I/O	TEST USART BUS 3
TST BUS 02	79	I/O	TEST USART BUS 2
TST BUS 01	80	I/O	TEST USART BUS 1
TST BUS 00	82	I/O	TEST USART BUS 0
IOC BUS 15	84	I/O	IOC BUS 15
IOC BUS 14	85	I/O	IOC BUS 14
IOC BUS 13	86	I/O	IOC BUS 13
IOC BUS 12	87	I/O	IOC BUS 12
IOC BUS 11	88	I/O	IOC BUS 11
IOC BUS 10	89	I/O	IOC BUS 10
IOC BUS 09	91	I/O	IOC BUS 9

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6.4.12 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOC BUS 08	92	I/O	IOC BUS 8
IOC BUS 07	93	I/O	IOC BUS 7
IOC BUS 06	94	I/O	IOC BUS 6
IOC BUS 05	95	I/O	IOC BUS 5
IOC BUS 04	96	I/O	IOC BUS 4
IOC BUS 03	97	I/O	IOC BUS 3
IOC BUS 02	98	I/O	IOC BUS 2
IOC BUS 01	99	I/O	IOC BUS 1
IOC BUS 00	100	I/O	IOC BUS 0

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N739)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,5,8, 14,20, 21,23, 24,27, 33,37, 42,44, 45,47, 49,50, 62,63, 65,66, 68	Frame	10	INF	MOhm	1-9
A2	Isolation, used to unused pins	C	5,8,14, 20,21,23, 24,27,33, 37,42,44, 45,47,49, 50,62,63, 65,66,68	10	INF	MOhm	1-9
A3-A22	Isolation, unused to unused pins	see note 4		10	INF	MOhm	1-9

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A22
Isolation measurement	NA	± 500 KOhm	A1-A22

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 5,8,14,20,21,23,24,27, 33,37,42,44,45,47,49,50,62,63,65,66,68.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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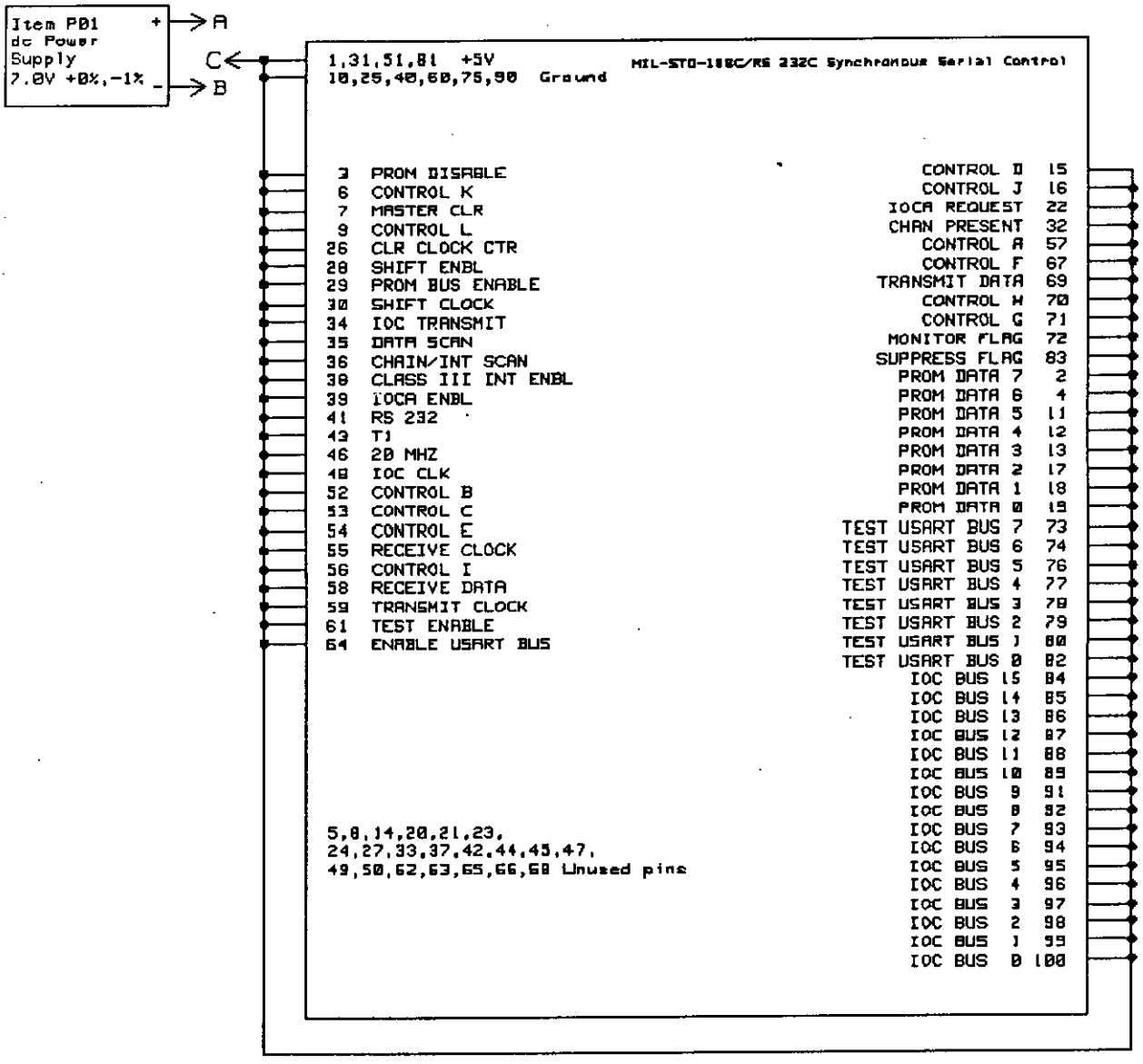


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, GND	10	60				
B6	Continuity, Vcc	1	51				
B7	Continuity, Vcc	51	81				
B8	Continuity, Vcc	1	51	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B8

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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MIL-STD-188C/RS 232C Synchronous Serial Control

1,31,51,81 +5V
10,25,40,60,75,90 Ground

3	PROM DISABLE	CONTROL D	15
5	CONTROL K	CONTROL J	16
7	MASTER CLR	IOCA REQUEST	22
9	CONTROL L	CHAN PRESENT	32
26	CLR CLOCK CTR	CONTROL A	57
28	SHIFT ENBL	CONTROL F	67
29	PROM BUS ENABLE	TRANSMIT DATA	69
30	SHIFT CLOCK	CONTROL H	70
34	IOC TRANSMIT	CONTROL G	71
35	DATA SCAN	MONITOR FLAG	72
36	CHAIN/INT SCAN	SUPPRESS FLAG	83
38	CLASS III INT ENBL	PROM DATA 7	2
39	IOCA ENBL	PROM DATA 6	4
41	RS 232	PROM DATA 5	11
43	T1	PROM DATA 4	12
46	20 MHZ	PROM DATA 3	13
48	IOC CLK	PROM DATA 2	17
52	CONTROL B	PROM DATA 1	18
53	CONTROL C	PROM DATA 0	19
54	CONTROL E	TEST USART BUS 7	73
55	RECEIVE CLOCK	TEST USART BUS 6	74
56	CONTROL I	TEST USART BUS 5	76
58	RECEIVE DATA	TEST USART BUS 4	77
59	TRANSMIT CLOCK	TEST USART BUS 3	78
61	TEST ENABLE	TEST USART BUS 2	79
64	ENABLE USART BUS	TEST USART BUS 1	80
		TEST USART BUS 0	82
		IOC BUS 15	84
		IOC BUS 14	85
		IOC BUS 13	86
		IOC BUS 12	87
		IOC BUS 11	88
		IOC BUS 10	89
		IOC BUS 9	91
		IOC BUS 8	92
5,8,14,20,21,23,		IOC BUS 7	93
24,27,33,37,42,44,45,47,		IOC BUS 6	94
49,50,62,63,65,66,68 Unused pins		IOC BUS 5	95
		IOC BUS 4	96
		IOC BUS 3	97
		IOC BUS 2	98
		IOC BUS 1	99
		IOC BUS 0	100

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.23 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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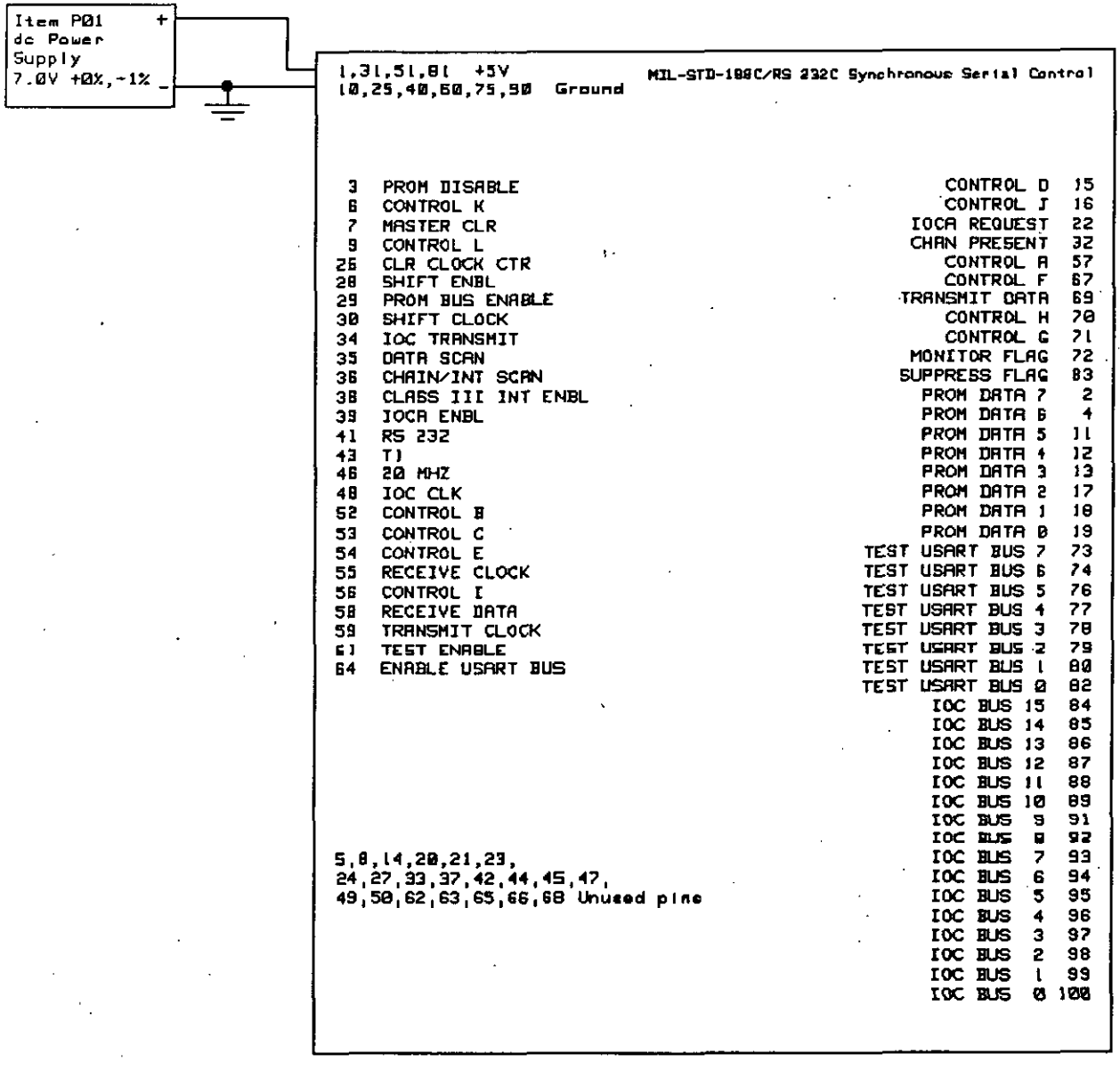


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	PROM DSABL	3	5.5	-0.12	1.11	mA	1-3
D2	CTRL K	6	5.5	0.0	0.1	mA	1-3
D3	MSTR CLR	7	5.5	0.0	0.2	mA	1-3
D4	CTRL L	9	5.5	0.0	0.1	mA	1-3
D5	CLR CLK	26	5.5	-0.12	1.21	mA	1-3
D6	SHIFT ENBL	28	5.5	-0.12	0.22	mA	1-3
D7	PROM BUS	29	5.5	0.0	0.1	mA	1-3
D8	SHIFT CLK	30	5.5	-0.12	0.22	mA	1-3
D9	IOC XMT	34	5.5	0.0	0.1	mA	1-3
D10	DATA SCAN	35					
D11	CHAIN/INT	36					
D12	C 3 INT	38					
D13	IOCA ENBL	39					
D14	RS 232	41					
D15	T1	43	5.5	0.0	0.1	mA	1-3
D16	20 MHZ	46	5.5	0.0	1.0	mA	1-3
D17	IOC CLK	48	5.5	0.0	4.0	mA	1-3
D18	CTRL B	52	5.5	0.0	0.1	mA	1-3
D19	CTRL C	53					
D20	CTRL E	54	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	RCV CLK	55	5.5	0.0	0.1	mA	1-3
D22	CTRL I	56					
D23	RCV DATA	58					
D24	XMT CLK	59	5.5	0.0	0.1	mA	1-3
D25	TEST ENBL	61	5.5	-0.12	0.32	mA	1-3
D26	ENBL USART	64	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D26
Power supply current limit	1.86 A	NA	D1-D26
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D26
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D26 D1-D26
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 61.5 uA ± 5.0 uA ± 10.0 uA ± 5.0 uA ± 66.5 uA ± 17.0 uA ± 5.0 uA ± 17.0 uA ± 5.0 uA ± 50.0 uA ± 200.0 uA ± 5.0 uA ± 22.0 uA ± 5.0 uA	D1 D2 D3 D4 D5 D6 D7 D8 D9-D15 D16 D17 D18-D24 D25 D26

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	DATA 07	2	0.0	-225.0	-40.0	mA	1-4
E2	DATA 06	4					
E3	DATA 05	11					
E4	DATA 04	12					
E5	DATA 03	13	0.0	-225.0	-40.0	mA	1-4
E6	CTRL D	15	0.0	-220.0	-30.0	mA	1-4
E7	CTRL J	16	0.0	-220.0	-30.0	mA	1-4
E8	DATA 02	17	0.0	-225.0	-40.0	mA	1-4
E9	DATA 01	18					
E10	DATA 00	19	0.0	-225.0	-40.0	mA	1-4
E11	IOCA RQST	22	0.0	-150.0	-60.0	mA	1-4
E12	CTRL A	57	0.0	-220.0	-30.0	mA	1-4
E13	CTRL F	67					
E14	XMT DATA	69					
E15	CTRL H	70					
E16	CTRL G	71	0.0	-220.0	-30.0	mA	1-4
E17	TST BUS 07	73	0.0	-225.0	-40.0	mA	1-4
E18	TST BUS 06	74					
E19	TST BUS 05	76					
E20	TST BUS 04	77	0.0	-225.0	-40.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	TST BUS 03	78	0.0	-225.0	-40.0	mA	1-4
E22	TST BUS 02	79					
E23	TST BUS 01	80					
E24	TST BUS 00	82	0.0	-225.0	-40.0	mA	1-4
E25	IOC BUS 15	84	0.0	-150.0	-60.0	mA	1-4
E26	IOC BUS 14	85					
E27	IOC BUS 13	86					
E28	IOC BUS 12	87					
E29	IOC BUS 11	88					
E30	IOC BUS 10	89					
E31	IOC BUS 09	91					
E32	IOC BUS 08	92					
E33	IOC BUS 07	93					
E34	IOC BUS 06	94					
E35	IOC BUS 05	95					
E36	IOC BUS 04	96					
E37	IOC BUS 03	97					
E38	IOC BUS 02	98					
E39	IOC BUS 01	99					
E40	IOC BUS 00	100	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E40
Power supply current limit	1.86 A	NA	E1-E40
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E40
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E40 E1-E40
Output current measurement	NA NA NA NA NA NA NA	± 9.25 mA ± 9.5 mA ± 9.25 mA ± 4.5 mA ± 9.5 mA ± 9.25 mA ± 4.5 mA	E1-E5 E6,E7 E8-E10 E11 E12-E16 E17-E24 E25-E40

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	DATA 07	2	2.4	-0.2	0.02	mA	1-3
F2	PROM DSABL	3	2.4	-5.68	-2.86	mA	1-3
F3	DATA 06	4	2.4	-0.2	0.02	mA	1-3
F4	CTRL K	6	2.4	-0.2	0.02	mA	1-3
F5	MSTR CLR	7	2.4	-0.4	0.04	mA	1-3
F6	CTRL L	9	2.4	-0.2	0.02	mA	1-3
F7	DATA 05	11					
F8	DATA 04	12					
F9	DATA 03	13					
F10	DATA 02	17					
F11	DATA 01	18					
F12	DATA 00	19	2.4	-0.2	0.02	mA	1-3
F13	CLR CLK	26	2.4	-12.7	-2.75	mA	1-3
F14	SHIFT ENBL	28	2.4	-6.03	-2.85	mA	1-3
F15	PROM BUS	29	2.4	-0.2	0.02	mA	1-3
F16	SHIFT CLK	30	2.4	-6.03	-2.85	mA	1-3
F17	IOC XMT	34	2.4	-0.6	0.02	mA	1-3
F18	DATA SCAN	35					
F19	CHAIN/INT	36					
F20	C 3 INT	38	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	IOCA ENBL	39	2.4	-0.6	0.02	mA	1-3
F22	RS 232	41	2.4	-0.2	0.02	mA	1-3
F23	T1	43	2.4	-0.6	0.02	mA	1-3
F24	20 MHZ	46	2.4	-2.0	0.05	mA	1-3
F25	IOC CLK	48	2.4	-8.0	0.2	mA	1-3
F26	CTRL B	52	2.4	-0.2	0.02	mA	1-3
F27	CTRL C	53					
F28	CTRL E	54					
F29	RCV CLK	55					
F30	CTRL I	56					
F31	RCV DATA	58					
F32	XMT CLK	59	2.4	-0.2	0.02	mA	1-3
F33	TEST ENBL	61	2.4	-5.83	-2.83	mA	1-3
F34	ENBL USART	64	2.4	-0.2	0.02	mA	1-3
F35	TST BUS 07	73					
F36	TST BUS 06	74					
F37	TST BUS 05	76					
F38	TST BUS 04	77					
F39	TST BUS 03	78					
F40	TST BUS 02	79	2.4	-0.2	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F41	TST BUS 01	80	2.4	-0.2	0.02	mA	1-3
F42	TST BUS 00	82	2.4	-0.2	0.02	mA	1-3
F43	IOC BUS 15	84	2.4	-0.25	0.07	mA	1-3
F44	IOC BUS 14	85					
F45	IOC BUS 13	86					
F46	IOC BUS 12	87					
F47	IOC BUS 11	88					
F48	IOC BUS 10	89					
F49	IOC BUS 09	91					
F50	IOC BUS 08	92					
F51	IOC BUS 07	93					
F52	IOC BUS 06	94					
F53	IOC BUS 05	95					
F54	IOC BUS 04	96					
F55	IOC BUS 03	97					
F56	IOC BUS 02	98					
F57	IOC BUS 01	99					
F58	IOC BUS 00	100	2.4	-0.25	0.07	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F58
Power supply current limit	1.86 A	NA	F1-F58
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F58
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F58 F1-F58
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 11 uA ± 141 uA ± 11 uA ± 22 uA ± 11 uA ± 500 uA ± 159 uA ± 11 uA ± 159 uA ± 31 uA ± 11 uA ± 31 uA ± 102 uA ± 410 uA ± 11 uA ± 150 uA ± 11 uA ± 16 uA	F1 F2 F3,F4 F5 F6-F12 F13 F14 F15 F16 F17-F21 F22 F23 F24 F25 F26-F32 F33 F34-F42 F43-F58

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	DATA 07	2	0.4	-0.2	0.02	mA	1-3
G2	PROM DSABL	3	0.4	-5.68	-2.86	mA	1-3
G3	DATA 06	4	0.4	-0.2	0.02	mA	1-3
G4	CTRL K	6	0.4	-0.2	0.02	mA	1-3
G5	MSTR CLR	7	0.4	-0.4	0.04	mA	1-3
G6	CTRL L	9	0.4	-0.2	0.02	mA	1-3
G7	DATA 05	11					
G8	DATA 04	12					
G9	DATA 03	13					
G10	DATA 02	17					
G11	DATA 01	18					
G12	DATA 00	19	0.4	-0.2	0.02	mA	1-3
G13	CLR CLK	26	0.4	-12.7	-2.75	mA	1-3
G14	SHIFT ENBL	28	0.4	-6.03	-2.85	mA	1-3
G15	PROM BUS	29	0.4	-0.2	0.02	mA	1-3
G16	SHIFT CLK	30	0.4	-6.03	-2.85	mA	1-3
G17	IOC XMT	34	0.4	-0.6	0.02	mA	1-3
G18	DATA SCAN	35					
G19	CHAIN/INT	36					
G20	C 3 INT	38	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	IOCA ENBL	39	0.4	-0.6	0.02	mA	1-3
G22	RS 232	41	0.4	-0.2	0.02	mA	1-3
G23	T1	43	0.4	-0.6	0.02	mA	1-3
G24	20 MHZ	46	0.4	-2.0	0.05	mA	1-3
G25	IOC CLK	48	0.4	-8.0	0.2	mA	1-3
G26	CTRL B	52	0.4	-0.2	0.02	mA	1-3
G27	CTRL C	53					
G28	CTRL E	54					
G29	RCV CLK	55					
G30	CTRL I	56					
G31	RCV DATA	58					
G32	XMT CLK	59	0.4	-0.2	0.02	mA	1-3
G33	TEST ENBL	61	0.4	-5.83	-2.83	mA	1-3
G34	ENBL USART	64	0.4	-0.2	0.02	mA	1-3
G35	TST BUS 07	73					
G36	TST BUS 06	74					
G37	TST BUS 05	76					
G38	TST BUS 04	77					
G39	TST BUS 03	78					
G40	TST BUS 02	79	0.4	-0.2	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
G41	TST BUS 01	80	0.4	-0.2	0.02	mA	1-3
G42	TST BUS 00	82	0.4	-0.2	0.02	mA	1-3
G43	IOC BUS 15	84	0.4	-0.25	0.07	mA	1-3
G44	IOC BUS 14	85					
G45	IOC BUS 13	86					
G46	IOC BUS 12	87					
G47	IOC BUS 11	88					
G48	IOC BUS 10	89					
G49	IOC BUS 09	91					
G50	IOC BUS 08	92					
G51	IOC BUS 07	93					
G52	IOC BUS 06	94					
G53	IOC BUS 05	95					
G54	IOC BUS 04	96					
G55	IOC BUS 03	97					
G56	IOC BUS 02	98					
G57	IOC BUS 01	99					
G58	IOC BUS 00	100	0.4	-0.25	0.07	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G58
Power supply current limit	1.86 A	NA	G1-G58
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G58
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G58
low, (VL)	0.0 V	± 100 mV	G1-G58
Input current measurement	NA	± 11 uA	G1
	NA	± 141 uA	G2
	NA	± 11 uA	G3,G4
	NA	± 22 uA	G5
	NA	± 11 uA	G6-G12
	NA	± 500 uA	G13
	NA	± 159 uA	G14
	NA	± 11 uA	G15
	NA	± 159 uA	G16
	NA	± 31 uA	G17-G21
	NA	± 11 uA	G22
	NA	± 31 uA	G23
	NA	± 102 uA	G24
	NA	± 410 uA	G25
	NA	± 11 uA	G26-G32
	NA	± 150 uA	G33
	NA	± 11 uA	G34-G42
	NA	± 16 uA	G43-G58

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	DATA 07	2	-0.16	2.4	4.5	V	1-4
H2	DATA 06	4					
H3	DATA 05	11					
H4	DATA 04	12					
H5	DATA 03	13					
H6	CTRL D	15					
H7	CTRL J	16					
H8	DATA 02	17					
H9	DATA 01	18					
H10	DATA 00	19					
H11	IOCA RQST	22					
H12	CTRL A	57					
H13	CTRL F	67					
H14	XMT DATA	69					
H15	CTRL H	70					
H16	CTRL G	71					
H17	MNTR FLAG	72					
H18	TST BUS 07	73					
H19	TST BUS 06	74					
H20	TST BUS 05	76	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H21	TST BUS 04	77	-0.16	2.4	4.5	V	1-4
H22	TST BUS 03	78					
H23	TST BUS 02	79					
H24	TST BUS 01	80					
H25	TST BUS 00	82					
H26	SUPRS FLAG	83	-0.16	2.4	4.5	V	1-4
H27	IOC BUS 15	84	-0.4	2.4	4.5	V	1-4
H28	IOC BUS 14	85					
H29	IOC BUS 13	86					
H30	IOC BUS 12	87					
H31	IOC BUS 11	88					
H32	IOC BUS 10	89					
H33	IOC BUS 09	91					
H34	IOC BUS 08	92					
H35	IOC BUS 07	93					
H36	IOC BUS 06	94					
H37	IOC BUS 05	95					
H38	IOC BUS 04	96					
H39	IOC BUS 03	97					
H40	IOC BUS 02	98	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	IOC BUS 01	99	-0.4	2.4	4.5	V	1-4
H42	IOC BUS 00	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H42
Power supply current limit	1.86 A	NA	H1-H42
Forcing current, (Ioh)	-0.16 mA -0.4 mA	± 16 uA ± 40 uA	H1-H26 H27-H42
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H42 H1-H42
Output voltage measurement	NA	± 105 mV	H1-H42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

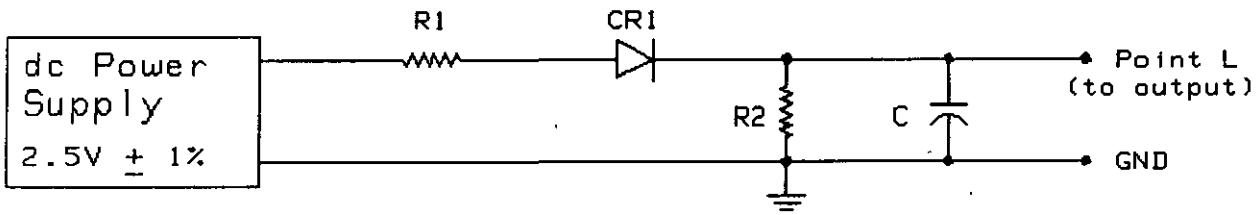
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,4,11,12,13,15, 16,17,18,19,22,32, 67,69,70,71,72,73, 74,76,77,78,79,80, 82,83	16000	1.6	0.5
84,85,86,87,88,89, 91,92,93,94,95,96, 97,98,99,100	6200	6.0	0.5
32	NA	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	DATA 07	2	1.6	0.0	0.5	V	1-4
J2	DATA 06	4					
J3	DATA 05	11					
J4	DATA 04	12					
J5	DATA 03	13					
J6	CTRL D	15					
J7	CTRL J	16					
J8	DATA 02	17					
J9	DATA 01	18					
J10	DATA 00	19					
J11	IOCA RQST	22					
J12	CHAN PRSNT	32					
J13	CTRL A	57					
J14	CTRL F	67					
J15	XMT DATA	69					
J16	CTRL H	70					
J17	CTRL G	71					
J18	MNTR FLAG	72					
J19	TST BUS 07	73					
J20	TST BUS 06	74	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	TST BUS 05	76	1.6	0.0	0.5	V	1-4
J22	TST BUS 04	77					
J23	TST BUS 03	78					
J24	TST BUS 02	79					
J25	TST BUS 01	80					
J26	TST BUS 00	82					
J27	SUPRS FLAG	83	1.6	0.0	0.5	V	1-4
J28	IOC BUS 15	84	6.0	0.0	0.5	V	1-4
J29	IOC BUS 14	85					
J30	IOC BUS 13	86					
J31	IOC BUS 12	87					
J32	IOC BUS 11	88					
J33	IOC BUS 10	89					
J34	IOC BUS 09	91					
J35	IOC BUS 08	92					
J36	IOC BUS 07	93					
J37	IOC BUS 06	94					
J38	IOC BUS 05	95					
J39	IOC BUS 04	96					
J40	IOC BUS 03	97	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	IOC BUS 02	98	6.0	0.0	0.5	V	1-4
J42	IOC BUS 01	99	″	″	″	″	″
J43	IOC BUS 00	100	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:31Z
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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J43
Power supply current limit	1.86 A	NA	J1-J43
Forcing current, (Iol)	1.6 mA 6.0 mA	± 160 uA ± 600 uA	J1-J27 J28-J43
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J43 J1-J43
Output voltage measurement	NA	± 25 mV	J1-J43

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K	CHAN PRSNT	32	5.5	0.0	1.0	mA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K
Power supply current limit	1.86 A	NA	K
Forcing voltage, (Voh)	5.5 V	± 50 mV	K
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K K
Output current measurement	NA	± 50 uA	K

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	CTRL K	6	-18.0	-1.5	0.0	V	1-3
L2	MSTR CLR	7					
L3	CTRL L	9					
L4	PROM BUS	29	-18.0	-1.5	0.0	V	1-3
L5	IOC XMT	34	-18.0	-1.2	0.0	V	1-3
L6	DATA SCAN	35					
L7	CHAIN/INT	36					
L8	C 3 INT	38					
L9	IOCA ENBL	39	-18.0	-1.2	0.0	V	1-3
L10	RS 232	41	-18.0	-1.5	0.0	V	1-3
L11	T1	43	-18.0	-1.2	0.0	V	1-3
L12	20 MHZ	46					
L13	IOC CLK	48	-18.0	-1.2	0.0	V	1-3
L14	CTRL B	52	-18.0	-1.5	0.0	V	1-3
L15	CTRL C	53					
L16	CTRL E	54					
L17	RCV CLK	55					
L18	CTRL I	56					
L19	RCV DATA	58					
L20	XMT CLK	59	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	ENBL USART	64	-18.0	-1.5	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L21
Power supply current limit	NA	NA	L1-L21
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L21
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L21 L1-L21
Input voltage measurement	NA NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1-L4 L5-L9 L10 L11-L13 L14-L21

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M	PSFC	1	6.6	12.4	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M
Capacitance measurement	NA	± 290 nF	M

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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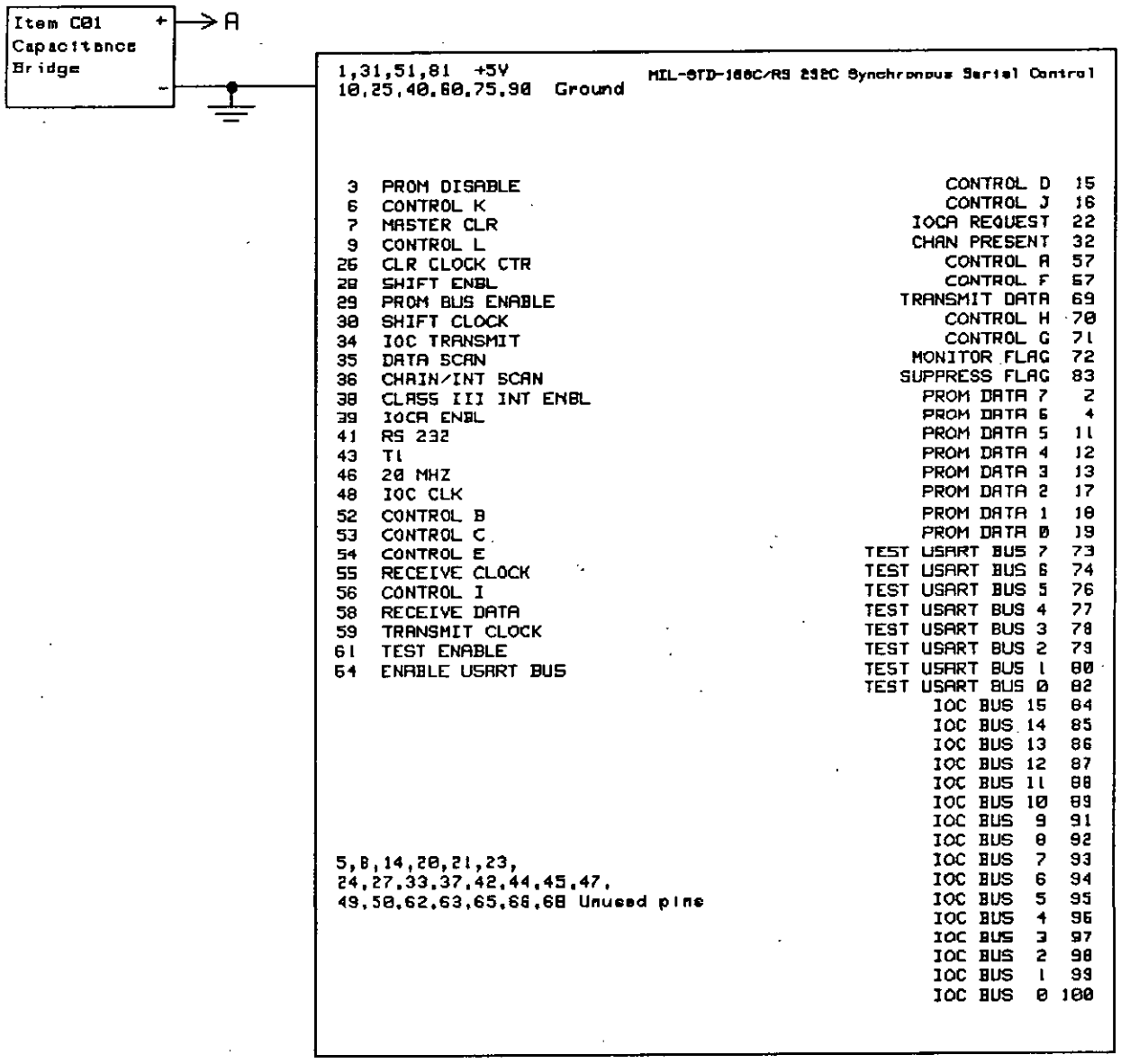


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	PROM DSABL	3	510	1240	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.86 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 36.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.86 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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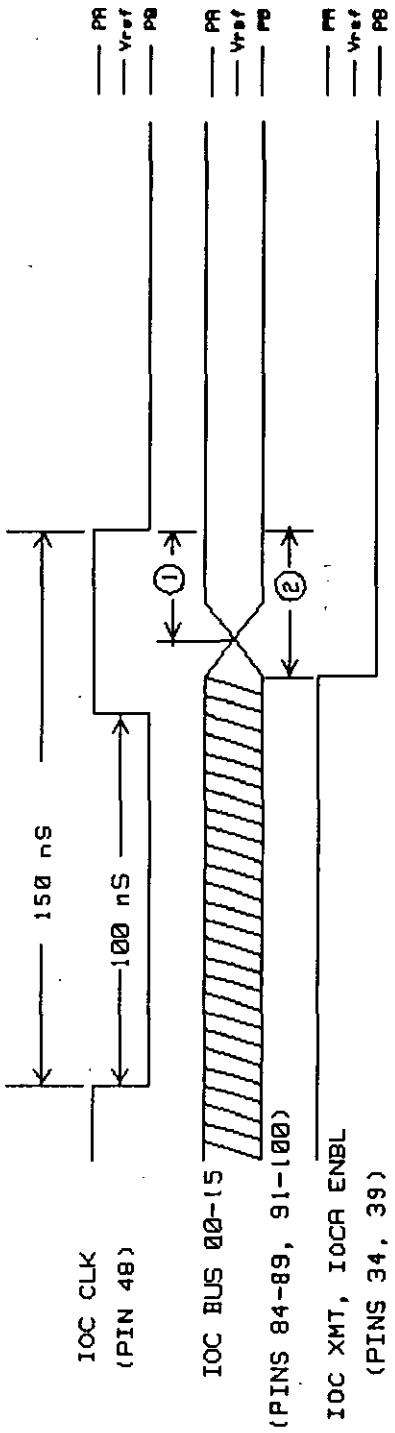


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) IOC BUS 00 - IOC BUS 15 to IOC CLK. Setup time: 24 nS minimum.

2. (2) IOC XMT and IOCA ENBL to IOC CLK. Setup time: 30 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.86 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

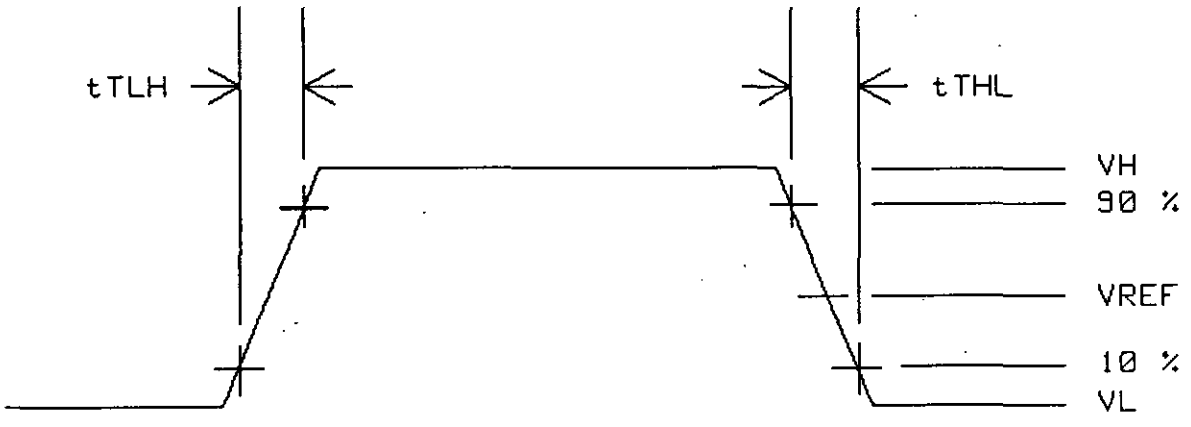
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	IOC CLK to IOC BUS 00	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 01	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 02	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 03	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 04	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 05	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 06	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 07	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 08	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 09	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 10	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 11	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 12	-	-	-	67	67	67	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	IOC CLK to IOC BUS 14	-	-	-	67	67	67	nS	12
1	tpd	IOC CLK to IOC BUS 15	-	-	-	67	67	67	nS	12
2	tsu	IOC BUS 00 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 01 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 02 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 03 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 04 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 05 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 06 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 07 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 08 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 09 to IOC CLK	24	24	24	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tsu	IOC BUS 10 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 11 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 12 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 13 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 14 to IOC CLK	24	24	24	-	-	-	nS	12
2	tsu	IOC BUS 15 to IOC CLK	24	24	24	-	-	-	nS	12
3	tsu	IOC XMT to IOC CLK	30	30	30	-	-	-	nS	12
3	tsu	IOCA ENBL to IOC CLK	30	30	30	-	-	-	nS	12
4	tpd	20 MHZ to DATA 00	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 01	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 02	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 03	-	-	-	250	250	250	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	20 MHZ to DATA 04	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 05	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 06	-	-	-	250	250	250	nS	12
4	tpd	20 MHZ to DATA 07	-	-	-	250	250	250	nS	12
5	tpd	20 MHz to TST BUS 00	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 01	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 02	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 03	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 04	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 05	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 06	-	-	-	192	192	192	nS	12
5	tpd	20 MHZ to TST BUS 07	-	-	-	192	192	192	nS	12
6	tpd	20 MHZ to SUPRS FLAG	-	-	-	81	81	81	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tpd	DATA SCAN to IOCA RQST	-	-	-	91	91	91	nS	12
8	tpd	TEST ENBL to DATA 00	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 01	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 02	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 03	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 04	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 05	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 06	-	-	-	52	52	52	nS	12
8	tpd	TEST ENBL to DATA 07	-	-	-	52	52	52	nS	12

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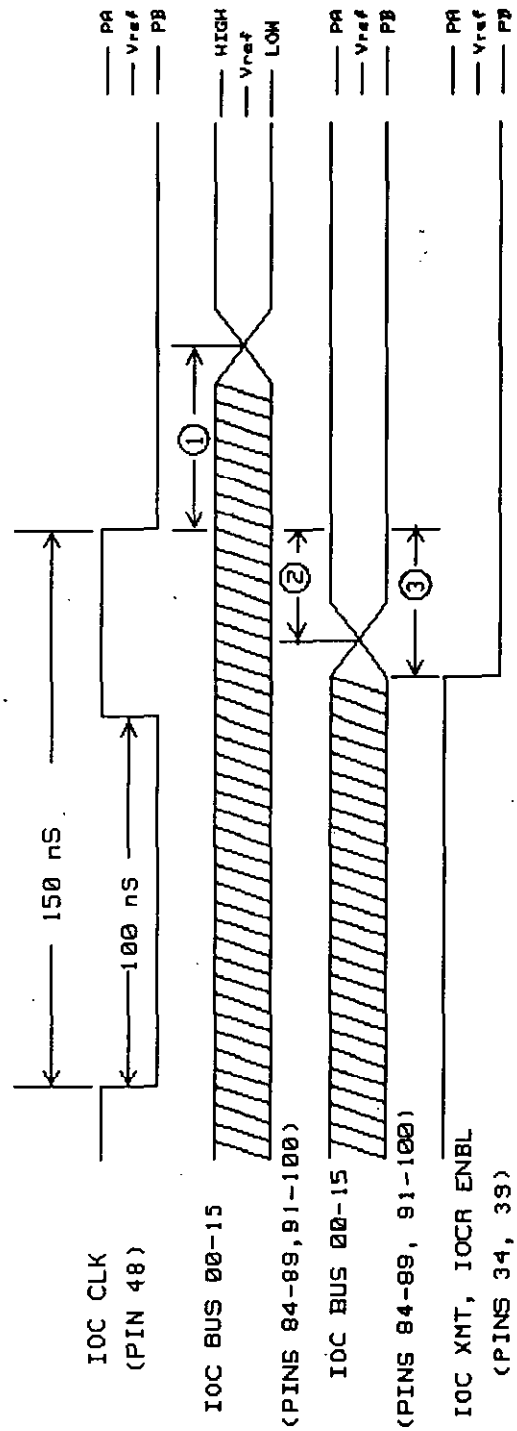


FIGURE 12a. Dynamic waveforms for table XVII.

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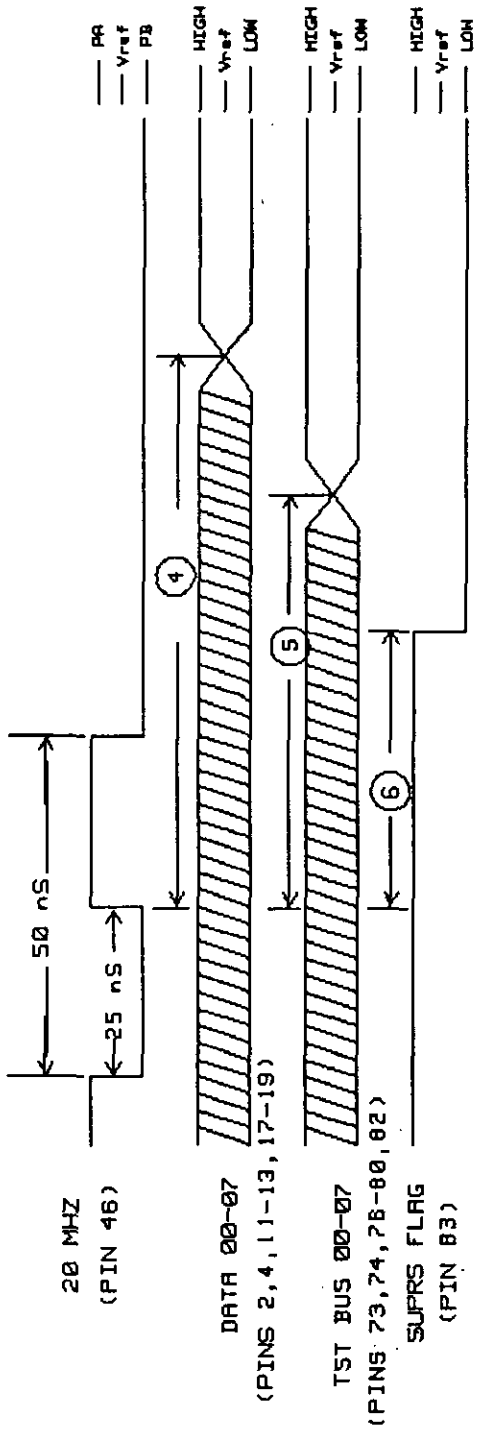


FIGURE 12b. Dynamic waveforms for table XVII.

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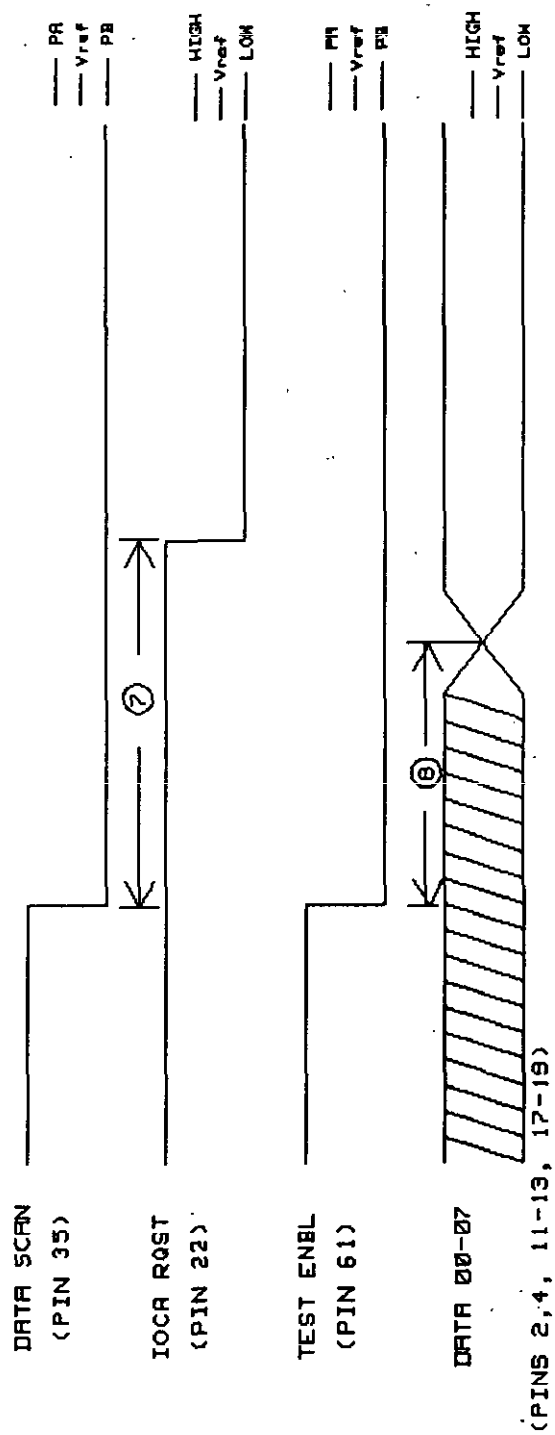


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	PROM DSABL	3	4	75	pF	1-4
R2	CTRL K	6	2	75	pF	1-4
R3	MSTR CLR	7	4	100	pF	1-4
R4	CTRL L	9	2	75	pF	1-4
R5	CLR CLK	26	6	100	pF	1-4
R6	SHIFT ENBL	28	2	75	pF	1-4
R7	PROM BUS	29	2	75	pF	1-4
R8	SHIFT CLK	30	2	100	pF	1-4
R9	IOC XMT	34	2	75	pF	1-4
R10	DATA SCAN	35	2	100	pF	1-4
R11	CHAIN/INT	36	2	75	pF	1-4
R12	C 3 INT	38				
R13	IOCA ENBL	39	2	75	pF	1-4
R14	RS 232	41	2	100	pF	1-4
R15	T1	43	2	75	pF	1-4
R16	20 MHZ	46	2	75	pF	1-4
R17	IOC CLK	48	8	100	pF	1-4
R18	CTRL B	52	2	75	pF	1-4
R19	CTRL C	53				
R20	CTRL E	54	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	RCV CLK	55	2	75	pF	1-4
R22	CTRL I	56	2	50	pF	1-4
R23	RCV DATA	58	2	75	pF	1-4
R24	XMT CLK	59	2	75	pF	1-4
R25	TEST ENBL	61	2	100	pF	1-4
R26	ENBL USART	64	2	75	pF	1-4
Bidirectionals						
R27	DATA 07	2	2	75	pF	1-4
R28	DATA 06	4				
R29	DATA 05	11				
R30	DATA 04	12				
R31	DATA 03	13				
R32	DATA 02	17				
R33	DATA 01	18				
R34	DATA 00	19				
R35	TST BUS 07	73				
R36	TST BUS 06	74				
R37	TST BUS 05	76				
R38	TST BUS 04	77	2	75	pF	1-4
R39	TST BUS 03	78	2	100	pF	1-4
R40	TST BUS 02	79	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	TST BUS 01	80	2	75	pF	1-4
R42	TST BUS 00	82	2	100	pF	1-4
R43	IOC BUS 15	84	3	75	pF	1-4
R44	IOC BUS 14	85				
R45	IOC BUS 13	86				
R46	IOC BUS 12	87				
R47	IOC BUS 11	88				
R48	IOC BUS 10	89				
R49	IOC BUS 09	91	3	75	pF	1-4
R50	IOC BUS 08	92	3	100	pF	1-4
R51	IOC BUS 07	93	3	100	pF	1-4
R52	IOC BUS 06	94	3	75	pF	1-4
R53	IOC BUS 05	95				
R54	IOC BUS 04	96				
R55	IOC BUS 03	97				
R56	IOC BUS 02	98				
R57	IOC BUS 01	99				
R58	IOC BUS 00	100	3	75	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R58
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R58
Capacitance measurement	NA	± 3.55 pF	R1
	NA	± 3.65 pF	R2
	NA	± 4.8 pF	R3
	NA	± 3.65 pF	R4
	NA	± 4.7 pF	R5
	NA	± 3.65 pF	R6,R7
	NA	± 4.9 pF	R8
	NA	± 3.65 pF	R9
	NA	± 4.9 pF	R10
	NA	± 3.65 pF	R11-R13
	NA	± 4.9 pF	R14
	NA	± 3.65 pF	R15,R16
	NA	± 4.6 pF	R17
	NA	± 3.65 pF	R18-R21
	NA	± 2.4 pF	R22
	NA	± 3.65 pF	R23,R24
	NA	± 4.9 pF	R25
	NA	± 3.65 pF	R26-R38
	NA	± 4.9 pF	R39
	NA	± 3.65 pF	R40,R41
	NA	± 4.9 pF	R42
	NA	± 3.6 pF	R43-R49
	NA	± 4.85 pF	R50,R51
	NA	± 3.6 pF	R52-R58

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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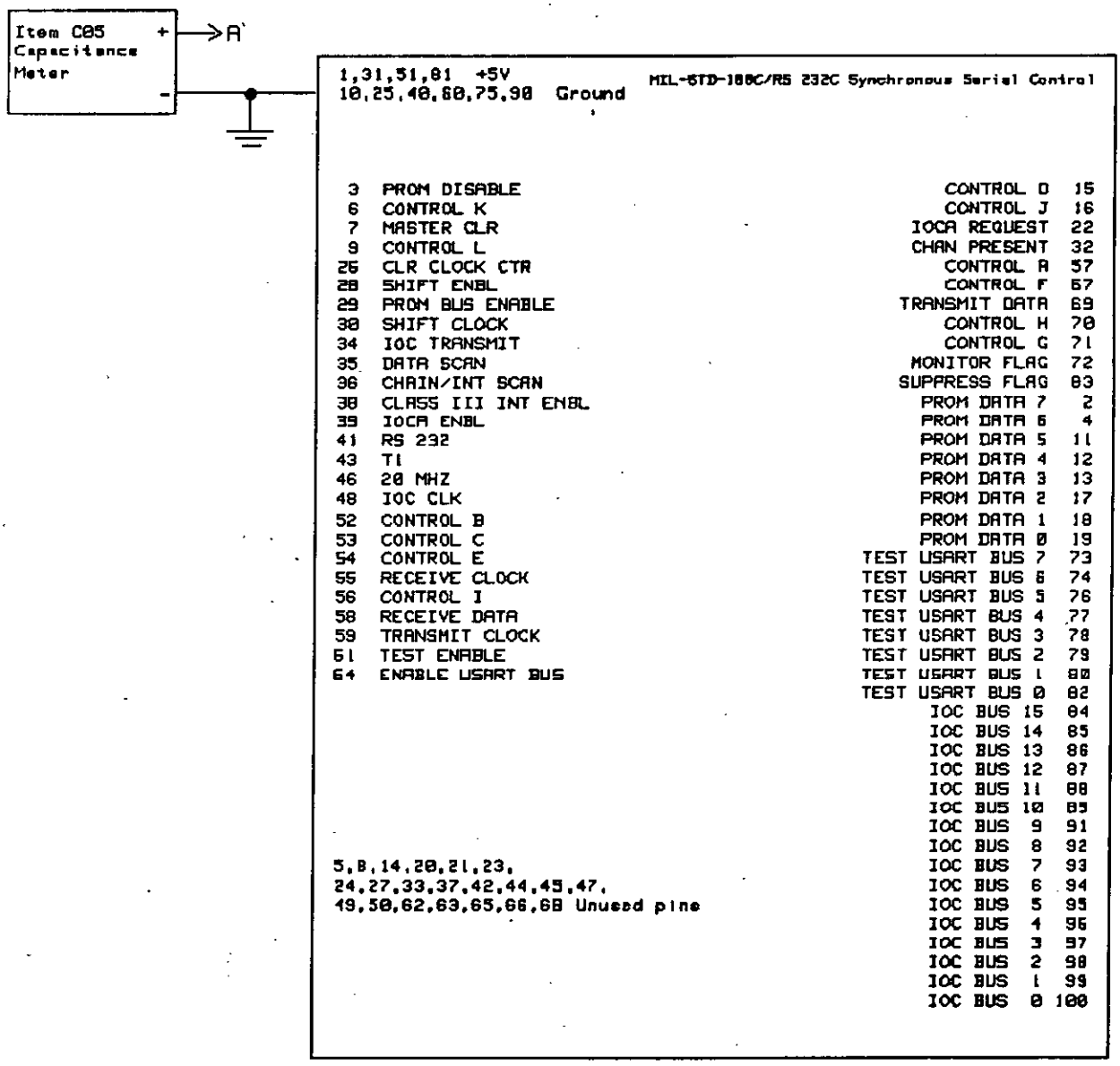


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T
Power supply current limit	1.86 A	NA	T
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T T
Power supply transient amplitude measurement	NA	± 20 mV	T

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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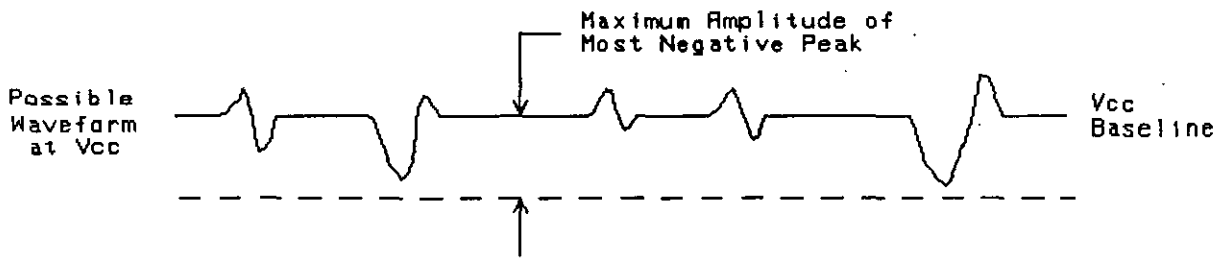
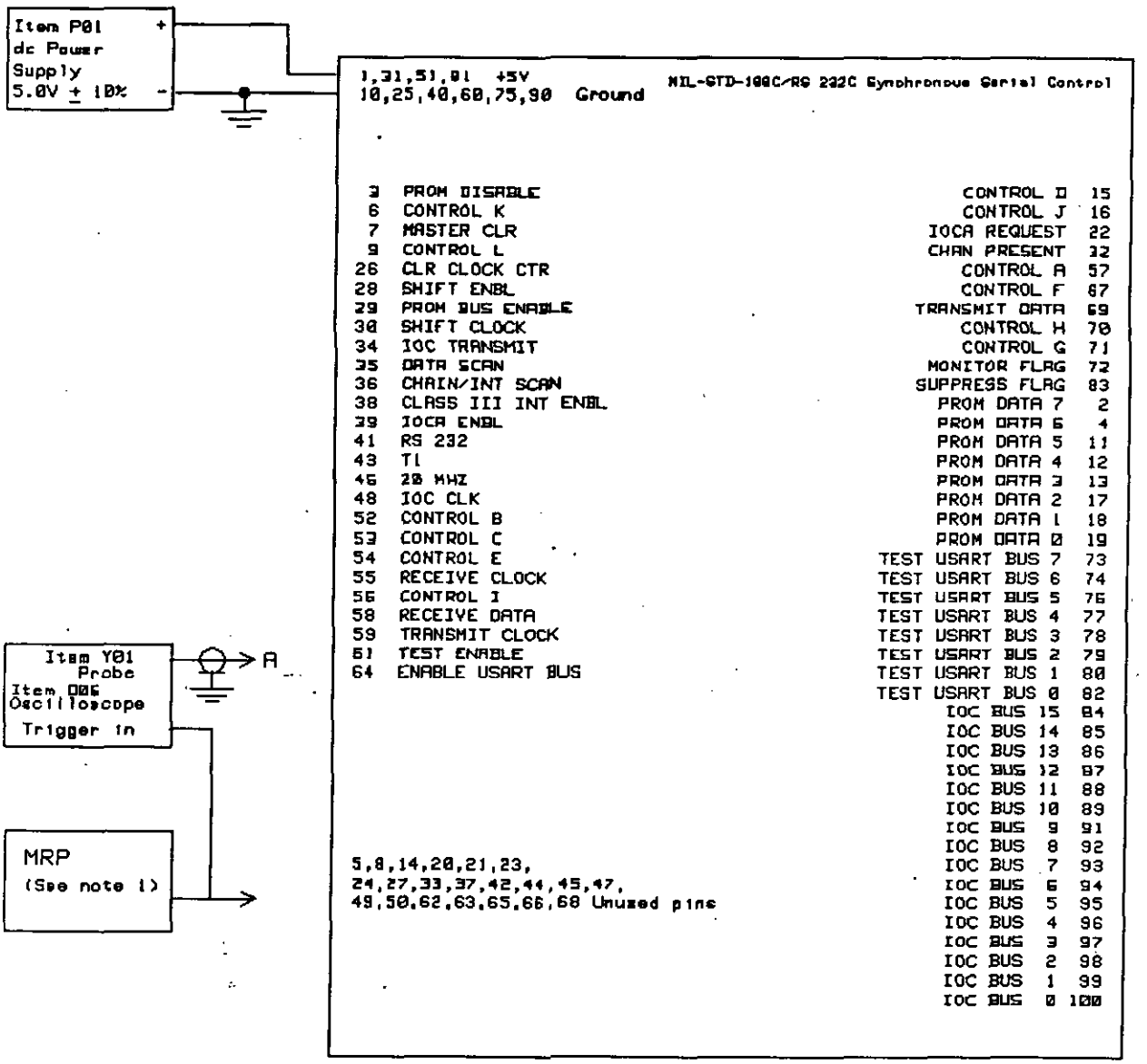


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/522	Modules, Standard Electronic, Synchronous Serial Control Module, Key Code AFK

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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